

Rishi Raj

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Education

Trident Academy of Technology (TAT), Bhubaneswar

2023 – 2027

B.Tech in Computer Science and Engineering (Data Science)

CGPA: 7.48/10

Experience

Database Intern – Zivanta Analytics (Remote)

Jun 2025 – Jul 2025

- Worked on **relational database operations** supporting structured data storage, retrieval, and reporting.
- Designed and maintained **relational tables** with proper keys and constraints for structured business data.
- Developed **PostgreSQL queries** using JOINS, subqueries, and aggregation to retrieve and transform data.
- Performed **data validation** and database maintenance to catch inconsistencies and integrity issues.

Data Analyst Intern – Mind Web Ventures (Remote)

Jun 2026 – Aug 2026

- Transformed **spatial, temporal, and business data** into analytical insights for data-backed decision-making.
- Analyzed **location-based and time-dependent datasets** to identify patterns for **locational intelligence**.
- Evaluated **business and industrial data** to find trends and risk patterns for **predictive analysis**.
- Prepared analytical outputs on **risk and ESG-related metrics**, turning raw data into business insights.

Projects

E-Commerce Customer Retention & CLV Analytics | [GitHub](#)

Oct 2024 – Dec 2024

- Built a **PostgreSQL**-based customer retention pipeline with **RFM segmentation** and **Power BI** reporting.
- Designed a **PostgreSQL schema** (8+ tables) for ~100K Olist orders, verifying row counts and integrity.
- Ran **RFM analysis** to segment customers into At Risk, Loyal, and Lost/Churned groups.
- Built a 2-page **Power BI dashboard**; surfaced a ~**3% repeat-purchase rate** as a key insight.

E-Commerce Review Intelligence Dashboard | [GitHub](#) & [Live Demo](#)

Feb 2025 – Apr 2025

- Built an **NLP-powered** review analytics dashboard to flag **rating-sentiment mismatches** for e-commerce brands.
- Scraped 977 Flipkart smartwatch reviews across 5 brands via **Selenium**; stored in **PostgreSQL** (Supabase).
- Applied a pretrained **DistilBERT** sentiment model; found ~**2x higher mismatch rate** for one brand.
- Delivered an interactive **Streamlit dashboard** with filters, comparisons, and automated GitHub Actions keep-alive.

Hospital Readmission Prediction | [GitHub](#) & [Live Demo](#)

Sep 2025 – Nov 2025

- Built a **healthcare ML model** to predict hospital readmission risk from patient records.
- Analyzed ~101K patient records (UCI Diabetes dataset, ~50 features) after cleaning.
- Engineered diagnosis/prior-visit features; identified **number_inpatient** as the top risk driver.
- Achieved ~**0.684 ROC-AUC**; deployed via **Flask API** and documented data ethics considerations.

Credit Risk & Loan Default Prediction | [GitHub](#) & [Live Demo](#)

Dec 2025 – Feb 2026

- Built an **end-to-end credit risk model** to predict loan default and deployed it as a **live API**.
- Cleaned ~150K borrower records (Kaggle); handled missing values, outliers, and sentinel codes.
- Engineered features and compared **Logistic Regression**, **Random Forest**, and **XGBoost** models.
- Reached ~**0.86 ROC-AUC** and ~**77% recall**; deployed via **Flask API** on Render.

Retail Sales Forecasting (Time Series) | [GitHub](#) & [Live Demo](#)

Mar 2026 – May 2026

- Built a **time-series forecasting system** comparing classical and ML models on 1M+ retail records.
- Analyzed ~1.02M daily sales records across 1,115 Rossmann stores (2013–2015) with store-level attributes.
- Engineered lag/rolling-window features with leakage-safe **TimeSeriesSplit** validation.
- Delivered ~**9.49% MAPE** with **XGBoost**, a ~**34% MAE improvement** over the SARIMA baseline.

Multi-Modal Transit & Logistics Route Optimization | [GitHub](#) & [Live Demo](#)

Sep 2026 – Oct 2026

- Built a **route optimization system** for a multi-vehicle delivery fleet using real road-network data.
- Extracted Indore's road network (~30K nodes) via **OSMnx**; solved a **CVRP** for a 4-vehicle fleet.
- Benchmarked **OR-Tools** vs. Nearest-Neighbor (**21.8% shorter routes**) and Dijkstra vs. **A*** (~3.25x faster).
- Visualized routes on a **Folium map**; deployed a live **Streamlit dashboard** with per-vehicle filters.

Technical Skills

Languages: Python, SQL

Data Analysis & Statistics: Data Cleaning, Exploratory Data Analysis (EDA), Statistics, RFM Analysis, Customer Segmentation, Time Series Analysis, Data Visualization

Databases & BI Tools: PostgreSQL, Microsoft Power BI, Microsoft Excel, Matplotlib, Seaborn

ML Libraries: Pandas, NumPy, Scikit-learn, XGBoost, LightGBM

Tools & Deployment: Jupyter Notebook, Git, GitHub, Flask, Streamlit, Render

Relevant Coursework

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|-----------------------|-----------------------|----------------------------------|--------------------|
| • Predictive Analysis | • Machine Learning | • Artificial Intelligence | • Computer Vision |
| • Statistics | • Database Management | • Data Mining & Data Warehousing | • Image Processing |

Position of Responsibility

Speech Weavers Club, Trident Academy of Technology

Jul 2024 – May 2025

Core Member

- Managed the club's social media presence across LinkedIn, Instagram, and Facebook, creating and publishing posts covering club activities, debates, and events for an audience of 500+ college students.
- Worked within the club's core team, coordinating on event promotion and regular audience communication alongside members handling other responsibilities.

Certifications

Google (Coursera), Mar 2026: Foundations of Data Science; The Power of Statistics; Regression Analysis: Simplify Complex Data Relationships; The Nuts and Bolts of Machine Learning

Knowledge Accelerators (Coursera), May 2026: Excel Basics for Beginners: Workbooks, Formatting, Charts and Core Skills